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Senior Software Engineer

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Summary

Senior Software Engineer specializing in **API** and backend **web services** solutions using various, well-known frameworks such as **Sanic**, **FastAPI**, Flask, Flask RESTful, **Django**, and Django REST API. Possessing deep, practical understanding of the **HTTP** protocol, web and network technologies, **AWS** GCP, and Linode cloud infrastructure, **Bash** and Zsh **scripting**, **Linux**, application security, and distributed version control tools such as **Git**. Highly experienced as a leader utilizing the **Agile** Methodology having successfully guided multiple **Scrum** teams to meet and exceed company goals.

Skills

- | | | |
|-------------------------|----------------------------------|-------------------------------|
| ▪ Python | ▪ Javascript | ▪ Node |
| ▪ Go | ▪ Bash | ▪ Zsh |
| ▪ Vim | ▪ Nvim | ▪ Git |
| ▪ Linux | ▪ API Development | ▪ Web backend |
| ▪ HTTP/HTTPS | ▪ AWS | ▪ GCP (Google Cloud Platform) |
| ▪ Docker | ▪ Kubernetes | ▪ Pandas |
| ▪ Flask & Flask RESTful | ▪ Django & Django REST Framework | ▪ Sanic |
| ▪ FastAPI | ▪ Agile | ▪ Scrum Methodology |
| ▪ Scrum Master | ▪ S.O.L.I.D. Design Principles | ▪ Clean Architecture |

Accomplishments

- Created code for ETL from/to various systems (Redis, PostgreSQL, Airtable, Jira, RESTful and GraphQL APIs, streaming services) for merging, sanitizing, updating, and pruning data and ensuring schema consistency.
- Set up pipelining for web services (APIs, applications) and event-driven ETL using AWS ECS, Fargate, Security Groups, and VPCs
- Leveraged Airflow as a more powerful alternative to CRON to manage event-driven and repeatable procedures in APIs, web backends, and ETL.
- Architected home construction business landing pages with email sequencing, SMS messaging, and payment processing.
- Used meta programming to create ORM interfacing multiple databases with Python objects.
- Worked with Pandas and Numpy to build Matplotlib charts of various types in service of financial reporting data and college thesis statistics.
- Designed API and frontend admin portal to manage shipping container business inventory logistics.

- Created ORM (Object Relational Mapper) leveraging Python descriptors to marshal database objects facilitating field type instantiation and customization.
- Created Docker containers utilizing Microservice model to deploy various backend web services in the AWS and GCP cloud environments for use as standalone solutions as well as dependency modules for API and web application code.
- Created custom MVC architecture around the Flask and Flask RESTful frameworks including a lightweight Object Relational Mapper for Python code objects to MongoDB.
- Developed and deployed a fully functional Python Flask RESTful API in support of a healthcare web application.
- Managed a team of overseas software engineers in the architecture, development, and deployment of a ReactJS healthcare web application portal.
- Deployed SSL-enabled web applications using Nginx, MongoDB on Linux Ubuntu.
- Developed code to send and receive email and SMS messages.
- Implemented custom Authentication and Authorization schemes for securing web APIs and applications.
- Keyed, installed, and configured SSL certificates in support of web applications and APIs served by Nginx.
- Developed, trained junior developers on the implementation of, and deployed custom Fernet encryption scheme to protect HTTP communication between ReactJS application and Python API.
- Architected a data modeling structure in support of a healthcare application allowing event driven communication flow.
- Implemented the Scrum Methodology, capturing business goals in the form of epics, user stories, and tasks which provided structure to development efforts resulting in a fully functional API and web application that attracted investors and clients.
- Developed a branching and rebasing policy document for junior developers to adhere to.
- Implemented marketing platform to send email and text messages to buyer contacts once a property is under contract
- Created seller questionnaire web page for potential sellers to sign up for real estate investment services.
- Created property sales web page to display properties under contract for purchase to real estate investor network.
- Set up the ability for marketing clients to log in to the web application and view detailed information for properties discovered via business marketing efforts.
- Set up the ability for partner wholesalers to log in and view properties assigned only to them for pursuit of sales opportunity.
- Created property analysis tool used to evaluate property offer amounts, and investor projected profit and ROI.

Experience

Senior Software Engineer

July 2023 to Present

Self-Employed

- Developing web applications and APIs for budgeting
- Developing web applications and APIs for church management operations
- Developing web applications and APIs for language translation
- Developing web applications and APIs for options and forex trading
- Preparing containerized domain modules using Docker

- Testing end-to-end functionality of web modules in docker-compose and KIND local dev environments
 - Configuring Redis and RabbitMQ servers for deployment in Docker/Kubernetes cloud environment
 - Configuring Kubernetes manifest files for application and API deployment over Linode Cloud
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Senior Software Engineer

May 2023 to July 2023

CVS, Inc. — (100% Remote)

- Rework legacy monolith code base in preparation for domain-driven, modular architecture
 - Prepare containerized domain modules using Docker and docker-compose
 - Advise management on domain-driven best practices
 - Test working monolithic code for concept efficacy with a view to modularization
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Senior Software Engineer

October 2021 to November 2022

Disney Corporation, Inc. — (100% Remote)

- Develop software for financial and reporting applications.
 - Collaborate with development team to design accounting costing system.
 - Configure and monitor cloud based architecture in support of web applications.
 - Implement functional and design changes in support of reporting web application maintenance.
 - Create unit tests in Java to ensure integrity of PostgreSQL functions
 - Created Python interface to PostgreSQL database facilitating CRUD operations
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Python Software Developer and System Architect

July 2019 to October 2021

CGJSoftware, Inc. — (100% remote)

- Test web applications and APIs using Python frameworks
- Create unit tests to ensure software integrity
- Deploy web application and RESTful API software
- Contribute to the smooth functioning of SDLC from conception to customer feedback
- Install SSL certs to secure web applications and RESTful API's
- Create site maps and routing diagrams for multiple applications
- Create charts to display application flow and provide development direction
- Lead team members in implementing functional areas of applications tied to url routing
- Containerize Python, React, and Node applications and API's for deployment in GCP
- Use multiple GCP services to run and launch consumer cloud web applications and API's
- Leverage GCP gcloud command line to
- Create and run SQL queries for use in Amazon AWS Redshift stress test

- Strong familiarity with AWS Python Boto3 package to gain programmatic access to AWS resources
- Creation and management of AWS EC2 instances for development and test environments
- Utilization of AWS Pub/Sub message to prevent application and microservice blocking and prevent data loss
- Responsible for architecture and program flow of client applications
- Developed flow charts as guidance for frontend developers application routing
- Created site map showing application routes available by version

API Developer, System Architect, DevOps Engineer, and Scrum Master

January 2018 to July 2019

Alan Health & Science Corp. — (100% remote)

- Architect software development and data modeling strategies to support business client needs
- Develop Python API software
- Review code commits checked in by overseas developers
- Develop pre-deploy check scripts to catch merging/rebasing conflicts and ReactJS coding errors Create database backup scripts using Bash and crontab
- Deploy Python API and ReactJS web application code to remote web servers Configure domain DNS settings for web server deployment
- Configure email domain TXT, CNAME, and MX records in DNS settings Install SSL certificates
- Setup and configure Linux Ubuntu server in preparation for web application and API deployment
- Give software development process guidance to business and development team members Train junior developers on Git branching and rebasing strategies

API Developer, Web Developer, and System Architect

July 2016 to January 2018

CT-REI, Inc. — Austin, TX

- Perform design, architect, implement, and maintain real estate investment and financial management web application
- Set up, configure, and maintain DNS for web application domain name
- Set up and maintain Nginx web servers on Google Cloud Compute instances for staging and production environments
- Set up and maintain MongoDB database server used to store web application data Write Python code to add to and improve features in the web application
- Perform SEO to increase page views for public facing web pages
- Test web application functionality and fix known issues in the application
- Write and maintain scripts to pull data from the Travis County Central Appraisal District website Write and maintain scripts to analyze business financial data

Quality Assurance Engineer

Sept 2016 to Oct 2016

Headway-Services — Austin, TX

Duties

- Design test plan.
- Convert manual test cases into automation test suites.
- Develop test scenarios that mimic user workflows.
- Business process and AGILE/SCRUM methodology consultation.

Accomplishments

- Communicated with Xojo desktop application via an open HTTP port within a test framework using the Python
 - requests package
 - Built test framework for automation testing using Python, Unittest, and Nosetests
 - Informed business on software testing architecture strategies
 - Educated team members on Scrum roles and methodology
 - Generated test cases out of product knowledge and feedback from customer UAT runs
 - Wrote scripts to parse xml documents for broken HTML tags inside CDATA enclosures
 - Edited XSLT code governing transformation of XML documents into HTML email templates
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Python Programmer

May 2016 to July 2016

Dell, Inc. — Austin, TX

Duties

- Refactor test framework software.
- Report weekly progress in standup meeting.
- Contribute Python development expertise to test framework project.

Accomplishments

- Removed sym-links across entire project eliminating circular file linkage
 - Removed duplicated code and modules from project.
 - Refactored core ConfigParser modules to be more object-oriented and intuitive. Fixed software defects in test framework.
 - Performed automation test runs using refactored test harness.
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Software Developer in Test

October 2015 to May 2016

Rackspace, Inc. — Austin, TX

Duties

- Create test protocol and write test cases for ticket routing engine and customer facing ticketing API
- Report plans and progress weekly on testing activities.
- Execute periodic automated test runs.
- Execute on-demand test runs to verify code readiness of ticketing component software
- Deliver verification sign-off for release of software deployment packages.

Accomplishments

- Created Python HTTP REST client used to execute tests against customer ticketing component software
 - Leveraged Object Oriented Programming principals to simplify request client API calls.
 - Provided verification that ticket routing engine and customer facing ticketing API operated as specified
 - Designed object oriented test-generator to create test cases via introspection of ticket routing engine rulesets.
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Software Developer and Scrum Master

April 2014 to August 2015

KeyIngredient — Austin, TX

Duties

- Guide Scrum team in process improvement as Scrum Master. Perform peer code reviews.
- Participate in bi-weekly stakeholders meetings and present new product ideas.
- Register and configure DNS domain names for applications hosted on Google AppEngine. Purchase and install SSL certificates for applications hosted on Google AppEngine.
- Create server-side API endpoints.
- Build and deploy e-commerce site functionality. Assist the team in the use of Git DVCS Apply frontend styling fixes to broken l18n translations.

Accomplishments

- Developed code to automate customer order fulfillment in compliance with ASC X12 EDI standards Built an email service provider comparison website.
 - Extracted and analyzed Google AppEngine datastore data Performed data analysis on customer spending trends Created email sending script and validation tool.
 - Integrated a cross-functional Scrum team providing guidance away from waterfall techniques Coached Scrum team in the implementation of a hierarchical ticketing system.
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Python API Developer

August 2013 to April 2014

Fidelity Investments — Irving, TX

Duties

- Supervise two junior developers, one on-site and one remote.
- Develop backend code using django rest framework.
- Develop frontend web application code using django framework Report on daily activities and progress in standup meetings Attend development activity weekly planning meetings.
- Attend daily backlog grooming meetings.

Accomplishments

- Mentored junior developers helping them to become more focused and productive datacenter.
 - Successfully delivered a fully functioning and tested web API that automated network device configuration for a corporate datacenter.
 - Successfully delivered a user interface which leveraged the API to allow users to make network configuration changes.
 - Gave valued input in meetings on the structure and layout of development stories.
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Quality Assurance and Network Integration Engineer

February 2005 to December 2009

Hospira, Inc. — San Diego, CA

Duties

- Validate various software tools used to assist on-market medical device management systems.
- Assist field service representatives in the installation, configuration, troubleshooting and proper use of database copying scripts.
- Build and configure windows 2003 server according to validated installation and configuration guidelines complete with relevant patches, hotfixes, and updates.
- Develop scripts to assist end users in the recovery of medical infusion device software in case of firmware upgrade failure.
- Develop software used to return medical device configuration to factory defaults.
- Recover and update software for medical infusion devices from an "unrecoverable state". Provide TCP/IP network support for embedded systems over wired and wireless networks.
- Setup, configure and maintain both wired and wireless LANs in support of team software verification efforts Update weekly engineering software builds and releases for formal verification and validation test runs. Perform weekly "smoke" test of individual software component builds and total build packages. Interface with development team to resolve complex software upgrade issues.
- Interface with management as a technical advisor on software upgrade issues.
- Install drug library management client and server side software and database packages for use with networked hospital infusion devices.
- Serve as verification team technical support contact to resolve any infusion device or personal computer networking issues.
- Execute requirements and session based manual testing.
- Submit detailed records of software defects in tracking database. Verify software defect fixes Formulate manual verification test plan.
- Write positive and negative test scenarios based upon given requirements.

Accomplishments

- Wrote a database load test tool used to stress a production database server in a successful effort to reproduce a customer complaint.
- Built, maintained, and tested enterprise level wireless security architecture from scratch utilizing various forms of the EAP protocol in conjunction with Cisco ACS server 4.0 and Linux FreeRadius.
- Installed and configured Linksys wireless network routers and Windows XP client machines with bot WEP and EAP level security.
- Installed, setup and configured Linux Suse 10 server to act as a FreeRadius EAP and DHCP network server Installed, setup and configured Cisco ACS 4.0 server in support

of EAP level wireless security network. Setup remote desktop to allow verification team users to connect to drug library server and client software. Wrote a Windows telnet client application with .NET 2.0 Framework using C++/CLI automatin.

- communication to hospital infusion devices using telnet.
- Participated in the software upgrade of 75 hospital infusion devices over two days in support of sales.
- marketing team building conference in Phoenix, AZ.
- Delivered medical device study second generation formal verification on time.
- Delivered five complete formal verification run-throughs for first-lot-to-stock ahead of schedule. Implemented verification strategy changes to dramatically improve testing efficiency.

Education

Associate of Science : Computer Electronics

2010 to 2013

ITT Technical Institute — San Diego, CA

Certificates

- Associate of Science Degree in Computer and Electronics Engineering Technology
- GPA 3.67

Microsoft and CompTIA Certifications : Computer Network Administration

2004 to 2005

Microskills Training Center — San Diego, CA

Certificates

- CompTIA A+ certification Network+ certification
- Microsoft Certified Professional in Managing a Windows 2000 Network Environment

Certificate of Completion : Video Cable Installation

October 1999

Cox Communications Federal Campus — San Diego, CA

Certificates

- Video Cable Installation Certification

Certification : Field Wire Installation

1995 to 1996

Marine Corps Air Ground Combat Center — San Diego, CA

Certificates

- Graduated from Field Wire Installation Course
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Basic Military Combat Skills : Military Basic Training

September 1995

Marine Corps Recruit Depot — San Diego, CA, USA

Awards

- Graduated from Basic Training
- Promoted to Private First Class